Stochastic Hydrology — Semester Project FS 2026

Impact of Glacial Lake Outburst Floods on Flood Frequency Statistics of the Simme River

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Abstract

Eight Glacial Lake Outburst Floods (GLOFs) from the Plaine Morte glacier (2011–2018) affected the Simme catchment before a supraglacial bypass channel ended the outburst regime in 2019. We test whether these events distort flood frequency statistics using two long-term discharge records at the Simme gauge in Lenk: daily maximum discharge (BAFU/FOEN, 1944–2025) and daily mean discharge (GRDC, 1944–2020). Under daily maxima, GLOFs produced the annual maximum in 4 of 8 GLOF years; under daily means in 6 of 8, because sustained outflows elevate the 24-hour average above brief storm peaks. The 2018 outburst ($\hat{T} \approx 789\,\mathrm{yr}$) dominates the statistical signal. Kolmogorov–Smirnov and Rank-Sum tests reject homogeneity between the GLOF-inclusive record and the pre-GLOF reference in both datasets ($p \leq 0.002$). After removing direct GLOF contributions with a $\pm$5-day masking window, both tests fail to reject homogeneity ($p \geq 0.17$), indicating the natural-flood regime remained stationary. A consistent but non-significant upward tendency in the masked peaks suggests possible gradual hydroclimatic change that warrants monitoring as post-bypass data accumulate. Naive inclusion of GLOF peaks overstates the 100-year design flood by approximately 15 %.

1 Introduction

Flood frequency analysis (FFA) rests on the assumption of stationarity: that annual maximum discharges are drawn from a single, time-invariant distribution. When a recurring process inflates extreme flows in a subset of years, this assumption is violated and fitted design quantiles become misleading.

The Simme river (canton Bern, Switzerland) provides a compelling case study. The Plaine Morte glacier hosts a supraglacial lake (Lac des Faverges) whose drainage produced eight documented Glacial Lake Outburst Floods (GLOFs) between 2011 and 2018. Since GLOFs are mechanistically distinct from the rainfall- and snowmelt-driven floods that dominate the pre-2011 record, their inclusion in a stationary FFA is conceptually problematic. Following the largest recorded GLOF in 2018, which caused approximately CHF 2.5 million in damage [GEOTEST AG, 2019], a supraglacial bypass channel was constructed in 2019 [GEOPRÆVENT AG, 2022], ending the uncontrolled outburst regime. The period 2011–2018 is therefore a closed historical episode whose statistical imprint on the long-term flood frequency record warrants retrospective investigation.

Both research questions are addressed in parallel using daily maximum instantaneous discharge (*Tagesmaxima*, BAFU 2219, 1944–2025) and daily mean discharge (GRDC 6935331, 1944–2020), making the choice of flow-aggregation statistic a key methodological variable.

2 Research Questions

1. Do recurring Glacial Lake Outburst Floods (GLOFs) from the Plaine Morte glacier systematically distort the flood frequency distribution of the Simme river at Lenk, and if so, by how much do GLOF-inclusive design quantiles exceed those from a homogeneous reference period?
2. Are the annual maximum discharges recorded during the GLOF period (2011–2018) statistically consistent with the pre-GLOF reference distribution once the direct GLOF contribution is removed, or does the post-2010 natural-flood regime show evidence of additional change?

3 Hypotheses

The research questions are operationalised as two testable statistical hypotheses.

Hypothesis 1 — GLOF inflation: GLOF-inclusive annual maxima from the 2011–2018 period are statistically incompatible with the pre-GLOF reference GEV distribution.

$H_0^{(1)}$: The 2011–2018 full annual maxima belong to the same population as the pre-GLOF reference period (no distributional shift).

$H_1^{(1)}$: The 2011–2018 full annual maxima differ from the pre-GLOF reference distribution (the GLOF signal shifts the tail).

Rejection of $H_0^{(1)}$ at $\alpha = 0.05$ confirms that the GLOF signal inflates the tail of the annual maximum series. *Tests:* KS test on GLOF-inclusive (2011–2018) vs. reference GEV fitted on 1944–2010 — direct test of $H_0^{(1)}$. Rank-Sum comparison 5a (pre-GLOF reference vs. GLOF-inclusive) directly tests $H_0^{(1)}$ non-parametrically. Comparison 5c (GLOF-inclusive vs. all non-GLOF years) provides additional corroboration using a broader reference group.

Hypothesis 2 — Regime stationarity: Once the direct GLOF contribution is removed by masking, the 2011–2018 annual maxima are statistically indistinguishable from the pre-GLOF reference, indicating that the underlying natural-flood regime remained stationary.

$H_0^{(2)}$: The 2011–2018 GLOF-masked annual maxima belong to the same population as the pre-GLOF reference period.

$H_1^{(2)}$: The 2011–2018 GLOF-masked annual maxima differ from the pre-GLOF reference distribution (evidence of additional hydroclimatic change beyond GLOFs).

Failure to reject $H_0^{(2)}$ at $\alpha = 0.05$ supports the stationarity of the natural-flood regime and attributes the observed distributional shift entirely to the direct GLOF signal. *Tests:* KS test on GLOF-masked (2011–2018) vs. reference GEV fitted on 1944–2010 and Rank-Sum comparison 5b (pre-GLOF reference vs. GLOF-masked) — both directly test $H_0^{(2)}$. Rank-Sum comparison 5d (GLOF-masked vs. all non-GLOF years) provides additional corroboration with a broader comparison group.

4 Data

Daily maximum discharge (*Tagesmaxima*) for the Simme at Oberried/Lenk (BAFU/FOEN station 2219) spans 1974–2025. Combined with *Monatsmaxima* for 1944–1973 from the same station, this yields an extended Annual Maximum Series (AMS) of $n = 82$ years (1944–2025). The monthly and daily sub-series agree exactly over the overlap period (maximum difference $= 0.000\,\mathrm{m^3/s}$), so the two sub-series are spliced at 1974 without bias. Both records were provided upon request by the Swiss Federal Office for the Environment [Swiss Federal Office for the Environment (BAFU/FOEN), 2026].

The supplementary analysis uses daily mean discharge from GRDC station 6935331 [Global Runoff Data Centre (GRDC), 2026] (Simme at Lenk, 1944–2020, $n = 77$), providing a 67-year pre-GLOF reference.

GLOF event dates were determined from the Lac des Faverges lake-level time series provided by Ogier et al. [2021], who instrumented the lake from 2012 to 2019 at 10-minute resolution (water level, inferred volume, computed discharge). The GLOF date is defined as the timestamp of peak lake outflow: maximum discharge during the outburst phase, identified by requiring (i) lake volume to be decreasing ($\Delta V < 0$) and (ii) computed lake discharge to exceed $5\,\mathrm{m^3/s}$. The 2011 event pre-dates the logger installation and its date is sourced from Gletschersee Lenk [2026]. Uncontrolled outbursts occurred in each year from 2011 to 2018; from 2019 a bypass channel eliminates spontaneous outbursts [GEOTEST AG, 2019]. The eight confirmed event dates are listed in Table 1.

Table 1: Confirmed GLOF event dates, Plaine Morte glacier (2011–2018). Dates are peak lake outflow from the Lac des Faverges time series; 2011 from Gletschersee Lenk [2026].

Year	GLOF date
2011	27 July 2011
2012	16 July 2012
2013	2 August 2013
2014	7 August 2014
2015	1 August 2015
2016	28 August 2016
2017	19 July 2017
2018	27 July 2018

5 Methods

5.1 Series Construction and GLOF Masking

Annual block maxima are extracted from each daily discharge record. Three annual maximum series (AMS) are constructed: (i) the **full** series spanning the complete record, (ii) a **GLOF-masked** series in which values within ±5 days of each confirmed GLOF date are set to missing before the block maximum is taken, and (iii) a **reference** series restricted to the pre-GLOF period (1944–2010). The five-day symmetric window captures the typical GLOF recession limb at this gauge. Hypothesis tests are applied exclusively to the GLOF period 2011–2018; post-2018 years are excluded because the bypass channel altered the outburst mechanism.

From these series, the test groups summarised in Table 2 are derived.

Table 2: Annual maximum series and test groups used in the homogeneity tests. GRDC data extend to 2020 only; n values for GRDC reflect this shorter record.

Label	Description	Period	n (BAFU 2219)	n (GRDC)
Reference	Pre-GLOF annual maxima; basis for GEV fit	1944–2010	67	67
Full AMS	Complete record, GLOF-inclusive	1944–2025	82	77
GLOF-masked AMS	Full AMS with ±5-day GLOF windows removed	1944–2025	82	77
GLOF-inclusive	Full annual maxima, GLOF period only	2011–2018	8	8
GLOF-masked	Masked annual maxima, GLOF period only	2011–2018	8	8
Non-GLOF years	All years outside 2011–2018	1944–2010, 2019–2025	74	69

5.2 GEV Flood Frequency Analysis

A Generalised Extreme Value (GEV) distribution is fitted to each AMS by maximum likelihood estimation. The GEV encompasses Gumbel ($\xi = 0$), Fréchet ($\xi > 0$), and Weibull ($\xi < 0$) families, allowing the data to determine the appropriate tail behaviour. Design quantiles for $T = 2, 5, 10, 20, 50, 100, 200\,\mathrm{yr}$ are derived from the fitted parameters with delta-method confidence intervals [Wilks, 2011]. Gringorten plotting positions are used for empirical return periods, and model fit is compared using the Akaike Information Criterion.

5.3 Homogeneity Tests

Two complementary tests compare the 2011–2018 annual maxima against the pre-GLOF reference, applied identically to both datasets.

KS test: each 2011–2018 peak is transformed to a uniform variate via the reference GEV CDF (probability integral transform). The Kolmogorov–Smirnov statistic tests departure from Uniform(0,1) with critical value $C_\alpha = K_\alpha/(\sqrt{n} + 0.12 + 0.11/\sqrt{n})$, $K_{0.05} = 1.358$ (Module 1, Eq. 3.3; Passalacqua 2026). Tests run separately for GLOF-inclusive ($n = 8$) and GLOF-masked ($n = 8$).

Rank-Sum test: the non-parametric Wilcoxon Rank-Sum test (Module 1, §2.4.1; Passalacqua 2026) runs four comparisons: (5a) pre-GLOF reference vs. GLOF-inclusive as a direct test of $H_0^{(1)}$, (5b) pre-GLOF reference vs. GLOF-masked as a direct test of $H_0^{(2)}$, (5c) GLOF-inclusive vs. all non-GLOF years as additional corroboration of $H_0^{(1)}$, and (5d) GLOF-masked vs. all non-GLOF years as additional corroboration of $H_0^{(2)}$.

6 Results

6.1 Descriptive Statistics

Table 3 summarises the BAFU 2219 annual maxima. GLOF years show a mean 70 % above non-GLOF years (29.35 m^3/s vs. 17.28 m^3/s), driven largely by the 2018 event. The relative gap is smaller in the GRDC daily-mean series (39 %), consistent with daily averaging dampening brief storm peaks.

Table 3: Descriptive statistics of the annual maximum series (1944–2025, BAFU 2219 extended AMS combining Monatsmaxima 1944–1973 and Tagesmaxima 1974–2025).

	Full series	Non-GLOF years	GLOF years (2011–2018)
n	82	74	8
Mean [m^3/s]	18.46	17.28	29.35
Std [m^3/s]	7.51	5.55	13.45
Median [m^3/s]	16.41	16.09	27.13
Min / Max [m^3/s]	9.89 / 56.90	9.89 / 34.67	15.55 / 56.90

6.2 GLOF Contribution to Annual Maxima

Table 4 reports the annual maximum, its implied return period under the reference GEV, and whether the ±5-day masking window captures the annual maximum for each GLOF year, for both discharge records.

Table 4: Annual maxima during the GLOF period 2011–2018 under daily maximum discharge (BAFU 2219) and daily mean discharge (GRDC). The implied return period $\hat{T}$ is computed under the respective pre-GLOF reference GEV (1944–2010 for both datasets). "GLOF = max?" indicates whether the ±5-day masking window contains the annual maximum.

		Daily max. (BAFU 2219)			Daily mean (GRDC)		
Year	GLOF date	Q_{ann} [m^3/s]	$\hat{T}$ [yr]	GLOF =max?	Q_{ann} [m^3/s]	$\hat{T}$ [yr]	GLOF =max?
2011	27 Jul	27.09	17.0	No	15.08	17.4	No
2012	16 Jul	19.12	3.8	Yes	14.56	13.6	Yes
2013	2 Aug	20.59	5.0	Yes	14.99	16.6	Yes
2014	7 Aug	40.48	126.7	No	21.09	220.0	Yes
2015	1 Aug	15.55	1.9	No	10.11	1.6	Yes
2016	28 Aug	27.16	17.2	Yes	13.97	10.2	No
2017	19 Jul	27.95	19.7	No	13.94	10.0	Yes
2018	27 Jul	56.90	788.8	Yes	19.24	107.1	Yes
GLOF = ann. max (of 8 GLOF yrs)			4 / 8			6 / 8	

Under daily maxima, the GLOF produced the annual maximum in 4 of 8 years (2012, 2013, 2016, 2018); in the remaining four, storm floods dominated, including a $\hat{T} \approx 127$-year event in 2014. Under daily means, GLOFs dominated in 6 of 8 years, because their sustained outflows elevate the 24-hour average above brief convective peaks.

6.3 GEV Parameter Estimates

Table 5 shows the MLE parameter estimates. Including GLOF years shifts $\hat{\xi}$ by +0.091 relative to the reference for BAFU 2219 and by +0.069 for GRDC; masking reduces this residual shift to +0.043 and −0.008 respectively, recovering near-reference tail behaviour. The AIC is substantially lower for the masked than the full-series fit.

Table 5: GEV MLE parameter estimates and AIC for both discharge datasets.

Dataset	Series	n	$\hat{\mu}$	$\hat{\sigma}$	$\hat{\xi}$	AIC
BAFU 2219 — daily maximum [m^3/s]						
	Full 1944–2025	82	14.94	3.95	0.251	513.66
	GLOF-masked 1944–2025	82	14.76	3.67	0.203	497.56
	Reference 1944–2010	67	14.52	3.55	0.160	399.73
GRDC — daily mean [m^3/s]						
	Full 1944–2020	77	10.211	1.711	0.170	346.93
	GLOF-masked 1944–2020	77	10.133	1.580	0.093	328.16
	Reference 1944–2010	67	10.038	1.542	0.101	283.54

6.4 Design Quantiles

Design quantiles are compared in Tables 6 and 7. The full AMS 100-year quantile exceeds the GLOF-masked estimate by 15.0 % for BAFU 2219 and 15.3 % for GRDC, growing to 18–20 % at 200 years. Masked and reference fits are closely aligned in both datasets.

Table 6: GEV design quantiles [m^3/s] for selected return periods — BAFU 2219 extended AMS (1944–2025), with 95 % delta-method confidence intervals [Wilks, 2011].

T [yr]	2	5	10	20	50	100	200
Full 1944–2025	16.47	22.12	26.81	32.38	41.08	48.99	58.68
GLOF-masked 1944–2025	16.17	21.19	25.18	29.75	36.61	42.61	49.71
Reference 1944–2010	15.87	20.53	24.09	28.04	33.77	38.61	44.17

Table 7: GEV design quantiles [m^3/s] for selected return periods — GRDC daily mean discharge.

T [yr]	2	5	10	20	50	100	200
Full 1944–2020	10.86	13.13	14.90	16.82	19.68	22.14	24.90
GLOF-masked 1944–2020	10.72	12.68	14.09	15.54	17.57	19.21	20.95
Reference 1944–2010	10.61	12.54	13.93	15.38	17.41	19.07	20.83

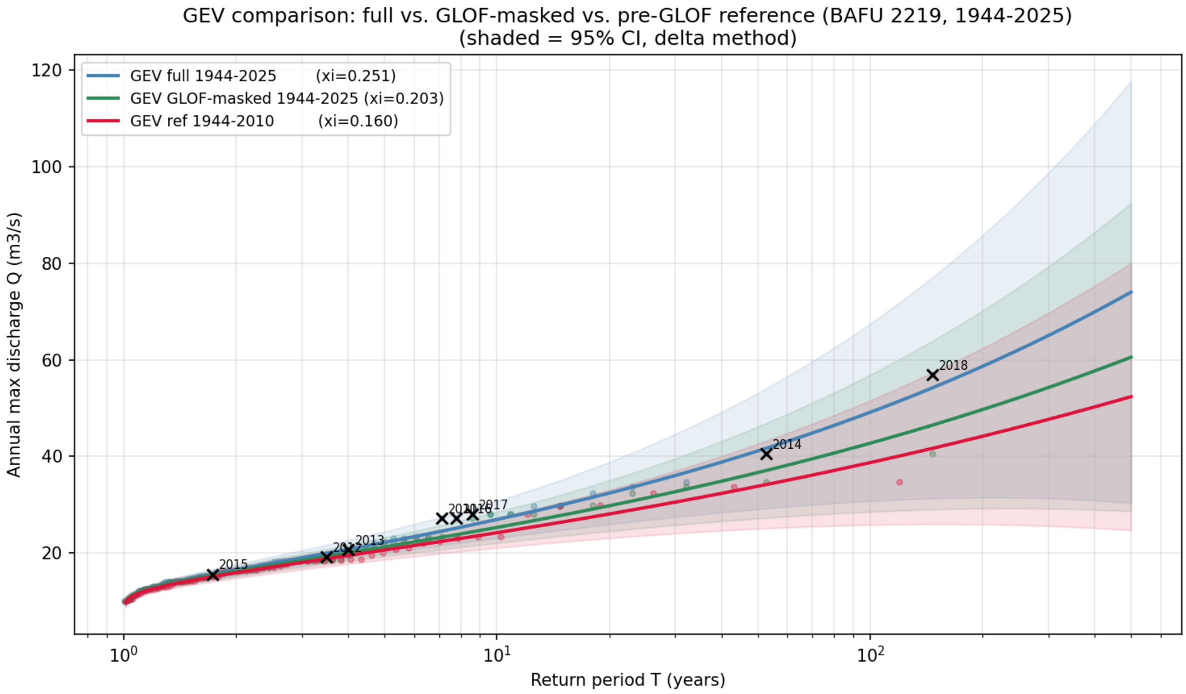

Figure 1: Return period plot for BAFU 2219 extended AMS (1944–2025): full (blue), GLOF-masked 1944–2025 (green), and reference 1944–2010 (red) GEV fits with 95 % delta-method confidence bands [Wilks, 2011]. Black crosses mark GLOF-year annual maxima from the full series with year labels. The 2018 event at $\hat{T} \approx 789$ yr dominates the upper tail. The GLOF-masked curve lies slightly above the reference across all return periods.

6.5 Homogeneity Tests

6.5.1 KS Test

Table 8 reports the KS statistics ($n = 8$, $C_{0.05} = 0.455$). The GLOF-inclusive reject $H_0^{(1)}$ in both datasets (BAFU 2219: $D_n = 0.610$, $p = 0.002$; GRDC: $D_n = 0.776$, $p < 0.001$). The GLOF-masked fail to reject $H_0^{(2)}$ in both ($D_n < 0.37$, $p > 0.19$).

Table 8: KS test results: 2011–2018 peaks vs. pre-GLOF reference GEV (1944–2010) for both discharge datasets ($n = 8$; $C_{0.05} = 0.455$).

Dataset	Group	D_n	$C_{0.05}$	p-value	Decision
BAFU 2219 (daily max.)	GLOF-inclusive	0.610	0.455	0.002	Reject $H_0^{(1)}$ at 5%
	GLOF-masked	0.360	0.455	0.195	Fail to reject $H_0^{(2)}$
GRDC (daily mean)	GLOF-inclusive	0.776	0.455	<0.001	Reject $H_0^{(1)}$ at 5%
	GLOF-masked	0.318	0.455	0.321	Fail to reject $H_0^{(2)}$

6.5.2 Rank-Sum Test

Table 9 shows the Rank-Sum results. Comparison 5a directly tests $H_0^{(1)}$ and rejects in both datasets (BAFU 2219: $p = 0.002$; GRDC: $p < 0.001$). Comparison 5b directly tests $H_0^{(2)}$ and fails to reject in both (BAFU 2219: $p = 0.172$; GRDC: $p = 0.233$). Comparisons 5c and 5d corroborate these findings using the broader non-GLOF reference group.

Table 9: Rank-Sum test results (two-sided, $\alpha = 0.05$) for both discharge datasets. Comparisons 5a and 5b directly test $H_0^{(1)}$ and $H_0^{(2)}$ against the pre-GLOF reference; comparisons 5c and 5d use all non-GLOF years as additional corroboration.

Dataset		Comparison	W	p-value	Decision
BAFU 2219	5a	Pre-GLOF reference vs. GLOF-inclusive	83.0	0.002	Reject $H_0^{(1)}$
	5b	Pre-GLOF reference vs. GLOF-masked	188.0	0.172	Fail to reject $H_0^{(2)}$
	5c	GLOF-inclusive vs. non-GLOF years	499.0	0.002	Reject
	5d	GLOF-masked vs. non-GLOF years	382.0	0.182	Fail to reject
GRDC	5a	Pre-GLOF reference vs. GLOF-inclusive	65.0	<0.001	Reject $H_0^{(1)}$
	5b	Pre-GLOF reference vs. GLOF-masked	198.0	0.233	Fail to reject $H_0^{(2)}$
	5c	GLOF-inclusive vs. non-GLOF years	486.0	<0.001	Reject
	5d	GLOF-masked vs. non-GLOF years	349.0	0.226	Fail to reject

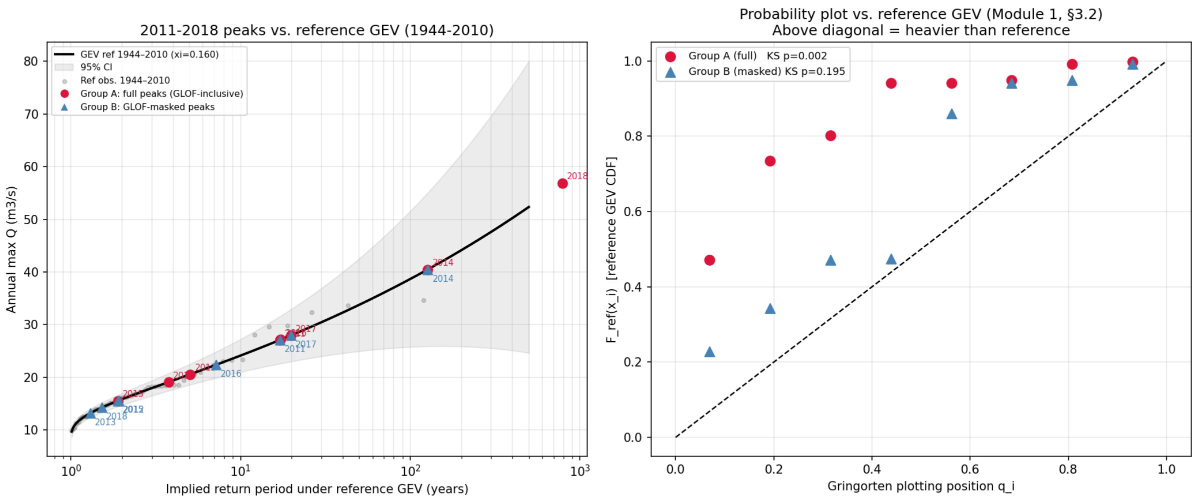

Figure 2: Left: 2011–2018 BAFU 2219 daily-maximum peaks on the reference GEV return-period curve (1944–2010). The 2018 GLOF-inclusive peak (crimson, $\hat{T} \approx 789\,\text{yr}$) dominates; after masking it drops to $\hat{T} \approx 1.5\,\text{yr}$ (blue triangle). Right: probability plot (Module 1, §3.2; Passalacqua 2026). GLOF-inclusive lie far above the diagonal, driven by 2018. GLOF-masked lie systematically above the 1:1 line, indicating peaks slightly higher than the reference predicts.

7 Discussion

Statistical signal driven by the 2018 event. The 2018 outburst ($Q = 56.90\,\text{m}^3/\text{s}$, $\hat{T} \approx 789\,\text{yr}$) is the primary driver of the KS rejection. The causal attribution rests on the masking experiment: after the ±5-day window, 2018's annual maximum collapses to $14.28\,\text{m}^3/\text{s}$ ($\hat{T} \approx 1.5\,\text{yr}$), confirming the extreme peak was attributable to the outburst, not concurrent meteorological forcing. With $n = 8$ the formal KS test carries limited inferential weight in isolation; the masking experiment provides the more compelling mechanistic evidence.

Hypothesis 1 confirmed. The KS test and Rank-Sum comparison 5a both reject $H_0^{(1)}$ in both datasets (BAFU 2219: $p = 0.002$; GRDC: $p < 0.001$). The stronger rejection for GRDC reflects GLOFs dominating 6 of 8 years under daily means versus 4 of 8 under daily maxima. The full AMS GEV overstates the 100-year design flood by 15.0 % for BAFU 2219 and 15.3 % for

GRDC relative to the GLOF-masked estimate, driven by an upward shift in the shape parameter ($\hat{\xi}$: 0.160 → 0.251 for BAFU 2219) and dominated by the 2018 event.

Hypothesis 2 supported. The KS test and Rank-Sum comparison 5b both fail to reject $H_0^{(2)}$ in both datasets (BAFU 2219: KS $p = 0.195$, Rank-Sum $p = 0.172$; GRDC: KS $p = 0.321$, Rank-Sum $p = 0.233$). Once GLOF contributions are removed, the natural-flood regime during 2011–2018 is statistically indistinguishable from the pre-GLOF reference. However, Figs. 1 and 2 reveal a consistent upward tendency: the masked GEV lies above the reference at all return periods and all eight GLOF-masked annual maxima plot above the reference diagonal. This could reflect gradual hydroclimatic change — for instance, increasing precipitation intensity under a warmer climate — but with $n = 8$ the tests cannot distinguish this from sampling variability. A longer post-bypass record is needed to assess this tendency.

Implications of the bypass intervention. The bypass channel constructed in 2019 [GEOTEST AG, 2019] has ended the uncontrolled GLOF episode. The GLOF-masked or pre-GLOF reference GEV is now the appropriate design distribution for natural flood hazard at Lenk. Naive inclusion of unmasked peaks overstates the 100-year design flood by 15 % and the 200-year flood by 18–20 %, leading to unnecessarily conservative standards in any post-bypass hazard reassessment.

Limitations. The principal limitation is sample size: only $n = 8$ GLOF-period years constrain the test groups, making results sensitive to a single outlier. Failure to reject $H_0^{(2)}$ cannot be interpreted as confirmation of stationarity. Future work should extend the analysis as post-bypass data accumulate and apply mixture models that explicitly separate the GLOF and natural-flood populations [Kite, 1977].

8 Conclusion

This study examines the influence of eight Plaine Morte GLOFs (2011–2018) on flood frequency statistics at the Simme gauge in Lenk, using daily maximum (BAFU 2219, 1944–2025, $n = 82$) and daily mean discharge (GRDC 6935331, 1944–2020). The key findings are:

1. **GLOF dominance depends critically on the discharge statistic.** Under daily maxima (BAFU 2219), the GLOF produced the annual maximum in 4 of 8 GLOF years; under daily means (GRDC), in 6 of 8, because sustained outflows elevate the daily average above brief storm peaks. The GLOF period also included an approximately 127-year natural storm flood (2014), two approximately 17–20-year events (2011, 2017), and one minor storm year (2015, $\hat{T}pprox 2\,\mathrm{yr}$).

2. **One extreme event dominates the signal.** The 2018 outburst ($Q = 56.90\,\mathrm{m^3/s}$, $\hat{T} \approx 789\,\mathrm{yr}$) drives the KS rejection. After masking, its return period collapses to $\hat{T} \approx 1.5\,\mathrm{yr}$, confirming the GLOF as the mechanistic cause.

3. **Tests reject homogeneity for GLOF-inclusive.** KS and Rank-Sum comparison 5a (pre-GLOF reference vs. GLOF-inclusive) both reject $H_0^{(1)}$ in both datasets (BAFU 2219: KS $p = 0.002$, Rank-Sum $p = 0.002$; GRDC: KS $p < 0.001$, Rank-Sum $p < 0.001$). With only $n = 8$ test observations, results are sensitive to a single extreme event.

4. **Masking restores homogeneity.** After the ±5-day mask, neither the KS test nor Rank-Sum comparison 5b (pre-GLOF reference vs. GLOF-masked, direct test of $H_0^{(2)}$) rejects in either dataset (BAFU 2219: KS $p = 0.195$, Rank-Sum $p = 0.172$; GRDC: KS $p = 0.321$, Rank-Sum $p = 0.233$). Comparison 5d (GLOF-masked vs. all non-GLOF years) corroborates (BAFU 2219: $p = 0.182$; GRDC: $p = 0.226$), confirming the natural-flood regime is statistically unchanged across periods.

5. **Design flood inflation is substantial but event-driven.** Unmasked GEV fitting overstates the 100-year design quantile by approximately 15–16 % in both datasets, driven by the upward shift in shape parameter ($\hat{\xi}: 0.160 \rightarrow 0.251$ for BAFU 2219).

For engineering applications, the GLOF-masked or pre-GLOF reference GEV is recommended as the design distribution for natural flood hazard at Lenk. The 2019 bypass channel makes this recommendation directly applicable to current planning.

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